

Name: **Solution**

MATH 241

SUMMER 2014

QUIZ #7

Problem 1.

Given the difference equation below,

$x[k] = \frac{5}{6}x[k-1] - \frac{1}{6}x[k-2] + f[k]$ with the initial conditions $x[-1] = x[-2] = -1$. The function $f[k]$ defined as,

$$f[k] = \begin{cases} 2^{-k} & \text{for } k \geq 0 \\ 0 & \text{for } k < 0 \end{cases}$$

- (7 Points) Find $x[k]$ for $k \geq 0$.
- (3 Points) Identify the homogeneous (zero input; $x_{ZI}[k]$) and the forced (zero state; $x_{ZS}[k]$) parts of the solution.
- (3 Points) Verify that $x_{ZI}[-1] = x[-1]$, $x_{ZI}[-2] = x[-2]$, $x_{ZS}[-1] = 0$ and $x_{ZS}[-2] = 0$.
- (2 Points) State and justify whether the system is stable or not.

$$(i) (ii) X(z) = \frac{5}{6} (z^{-1}X(z) + x[-1]) - \frac{1}{6} (z^{-2}X(z) + z^{-1}x[-1] + x[-2]) + F(z)$$

$$X(z) = \frac{5}{6} z^{-1}X(z) + \frac{5}{6} x[-1] - \frac{1}{6} z^{-2}X(z) - \frac{1}{6} z^{-1}x[-1] - \frac{1}{6} x[-2] + F(z)$$

$$X(z) \left(1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2} \right) = \frac{1}{6} (-x[-1]z^{-1} + 5x[-1] - x[-1]) + F(z)$$

$$X(z) = \frac{1}{6} \frac{-x[-1]z^{-1} + 5x[-1] - x[-1]}{\left(1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2} \right)} + \frac{F(z)}{\left(1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2} \right)}$$

only depends on the initial conditions.

$X_{ZI}(z)$

$X_{ZS}(z)$ only depends on input

Zero input solution, is the solution when we have no source i.e., $f[k]=0$
 - Zero state solution: The solution when we have zero initial conditions $x[-1]=x[-2]=0$

$$X_{ZI}(z) = \frac{1}{6} \left(\frac{z^{-1} - 4}{\left(1 - \frac{1}{2} z^{-1} \right) \left(1 - \frac{1}{3} z^{-1} \right)} \right) = \frac{A}{1 - \frac{1}{2} z^{-1}} + \frac{B}{1 - \frac{1}{3} z^{-1}} = \frac{1}{3 \left(1 - \frac{1}{3} z^{-1} \right)} - \frac{1}{\left(1 - \frac{1}{2} z^{-1} \right)}$$

$$A = \left(1 - \frac{1}{2} z^{-1} \right) X_{ZI}(z) \Big|_{z^{-1}=\frac{1}{2}} = \frac{1}{6} \frac{z^{-1} - 4}{1 - \frac{1}{3} z^{-1}} \Big|_{z^{-1}=\frac{1}{2}} = \frac{1}{6} \frac{-\frac{7}{2}}{1 - \frac{1}{3}} = -1$$

$$B = \left(1 - \frac{1}{3} z^{-1} \right) X_{ZI}(z) \Big|_{z^{-1}=\frac{1}{3}} = \frac{1}{6} \frac{z^{-1} - 4}{1 - \frac{1}{2} z^{-1}} \Big|_{z^{-1}=\frac{1}{3}} = \frac{1}{6} \frac{-\frac{11}{3}}{-\frac{1}{2}} = \frac{1}{3}$$

$$x_{ZI}[k] = \mathcal{Z}^{-1} \{ X_{ZI}(z^{-1}) \} = \mathcal{Z}^{-1} \left\{ \frac{1}{3} \frac{1}{1 - \frac{1}{3} z^{-1}} - \frac{1}{1 - \frac{1}{2} z^{-1}} \right\} = \frac{1}{3} \left(\frac{1}{3} \right)^k - \left(\frac{1}{2} \right)^k = \left(\frac{1}{3} \right)^{k+1} - \left(\frac{1}{2} \right)^k$$

Problem 2.

Given the differential equation below,

$x'' = -5x' - 6x + f(t)$ with the initial conditions $x(0) = 1, x'(0) = -1$. The function $f(t)$ is defined as,

$$f(t) = \begin{cases} e^{-2t} & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

- (7 Points) Find $x(t)$ for $t \geq 0$.
- (3 Points) Identify the homogeneous (zero input; $x_{ZI}(t)$) and the forced (zero state; $x_{ZS}(t)$) parts of the solution.
- (3 Points) Verify that $x_{ZI}(0) = x(0)$, $x'_{ZI}(0) = x'(0)$ and $x_{ZS}(0) = 0$, $x'_{ZS}(0) = 0$.
- (2 Points) State and justify whether the system is stable or not.

(i)(ii) $\mathcal{L}\{x''\} = \mathcal{L}\{-5x' - 6x + f(t)\}$

$$s^2 X(s) - sx(0) - x'(0) = -5(sX(s) - x(0)) - 6X(s) + F(s)$$

$$(s^2 + 5s + 6)X(s) = \underbrace{sx(0) + x'(0) + 5x(0)}_{\text{only depends on initial conditions!}} + \underbrace{F(s)}_{\text{only depends on input!}}$$

$$X(s) = \underbrace{\frac{sx(0) + x'(0) + 5x(0)}{(s+2)(s+3)}}_{X_{ZI}(s)} + \underbrace{\frac{F(s)}{(s+2)(s+3)}}_{X_{ZS}(s)} \rightarrow x(t) = x_{ZS}(t) + x_{ZI}(t)$$

$$X_{ZI}(s) = \frac{s+4}{(s+2)(s+3)} = \frac{A}{s+2} + \frac{B}{s+3} = \frac{6}{s+2} - \frac{1}{s+3} \xrightarrow{\mathcal{L}^{-1}} x_{ZI}(t) = 2e^{-2t} - e^{-3t}$$

$$A = (s+3)X_{ZI}(s) \Big|_{s=-2} = \frac{s+4}{s+3} \Big|_{s=-2} = 2 \quad B = (s+2)X_{ZI}(s) \Big|_{s=-3} = \frac{s+4}{s+2} \Big|_{s=-3} = -1$$

$$F(s) = \mathcal{L}\{e^{-2t}\} = \frac{1}{s+2}$$

$$X_{ZS}(s) = \frac{1/(s+2)}{(s+2)(s+3)} = \frac{1}{(s+2)^2(s+3)} = \frac{A}{s+2} + \frac{B}{(s+2)^2} + \frac{C}{s+3} = \frac{-1}{s+2} + \frac{1}{(s+2)^2} + \frac{1}{s+3} \xrightarrow{\mathcal{L}^{-1}} x_{ZS}(t) = -e^{-2t} + te^{-2t} + e^{-3t}$$

$$C = X_{ZS}(s)(s+3) \Big|_{s=-3} = \frac{1}{(s+2)^2} \Big|_{s=-3} = 1 \quad B = (s+2)^2 X_{ZS}(s) \Big|_{s=-2} = \frac{1}{s+3} \Big|_{s=-2} = 1$$

$$A = \frac{d}{dz} (s+2) \Big|_{s=-2} = \frac{d}{dz} (s+2)^2 X_{ZS}(s) \Big|_{s=-2} = \frac{d}{dz} \frac{1}{s+3} \Big|_{s=-2} = -\frac{1}{(s+3)^2} \Big|_{s=-2} = -1$$